

sockets, and estimates how long it takes for each pair to send data back and forth.

Hackbench was used with the following options:

```
$ hackbench -p -g 1 -f 2 -l 10000000 -s 1
```

How long Hackbench runs for is not known in advance; therefore, it is run with an arbitrarily large value of loops and forcefully terminated when Cyclictest finishes.

Latency with *kcompile* as load: To use *kcompile* as a load, we chose Rteval, which is written in Python to use two loads: Hackbench and a Linux kernel compile in periodic loops. While the loads are running, the Cyclictest program is run to measure system performance under these circumstances. After a specified duration, the loads are stopped and Cyclictest outputs are analysed by Rteval. We have run Rteval with just *kcompile* as load.

Future work

The above experiments and results have proved that the PREEMPT_RT patch with the GPOS Linux kernel can be used for hard real-time tasks. Linux is open source software published with the General Public License. Therefore, using Linux for hard real-time systems is cost-effective compared to using proprietary RTOS like VxWorks, etc. With this, we can eliminate the licence as well as royalty costs. So, the end product will be cheaper in a competitive market.

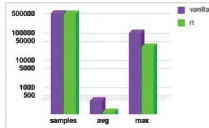


Figure 6: Latency results with Hackbench as load

Scheduling latency was the only parameter evaluated in this project. However, there are several other parameters like interrupt latency, I/O latency and network latency which also contribute to the deterministic behaviour of an RTOS. All of these can be measured and used to evaluate the performance of the vanilla kernel versus the real-time kernel. 

References

- [1] Stankovic, John A., and Raj Rajkumar: 'Real-time operating systems' *Real-Time Systems* 28.2-3 (2004): 237-253
- [2] Real-time Linux wiki. <https://rt.wiki.kernel.org/index.php/Cyclictest>
- [3] Felipe Cerqueira, Bjorn B. Brandenburg 'A Comparison of Scheduling Latency in Linux, PREEMPT RT, and LITMUSRT', Max Planck Institute for Software Systems (MPI-SWS)
- [4] Oliveira, Daniel Bristot, and Romulo Silva Oliveira: 'Timing analysis of the PREEMPT RT Linux kernel' *Software: Practice and Experience* 46.6 (2016): 789-819.
- [5] High resolution timers: https://rt.wiki.kernel.org/index.php/High_resolution_timers
- [6] Hackbench man page: <http://manpages.ubuntu.com/manpages/zesty/man8/hackbench.8.html>
- [7] Examining load average: <http://www.linuxjournal.com/article/9001>

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